Solutions — 5 Mock Exams · Data Science Introduction

This file contains the answer keys and short rationales for all five practice exams. For single-best-answer exams, one option is correct (1 point). For multiple-response exams, all listed options must be ticked (2 points; one mistake = 1 point; two or more mistakes = 0 points). Try each exam closed-book first, then check here.

Mock Exam 1 — Solutions & Answer Key (Single best answer (tick one) · 1 pt/question · 30 points)

Q	Topic	Answer	Rationale
1	1. VSM vs. System Dynamics	A	VSM (Beer) = recursive structure for viability; SD (Forrester) = simulation of behaviour over time. Both rely on feedback.
2	2. Ashby Adaptation	B	Adding capacity = amplification; focusing the portfolio = attenuation.
3	3. Data Science Distinction	D	A dataset is 'big' when its size is central to the problem; the other statements are false.
4	4. Real-Time Adaptive Models	C	Adaptive models must keep updating; C is incorrect.
5	5. Overfitting	A	Being too simple is underfitting, not overfitting; A is incorrect.
6	6. Visualization	B	A truncated axis is technically true but not fair - it can mislead the reader.
7	7. Complexity Examples	D	These traits are textbook examples of data complexity.
8	8. Class Weighting	B	Class weighting reweights the loss; SMOTE/sampling change the data, which class weighting does not.
9	9. Target-Oriented Action	D	Starting data-first and ignoring the goal is the failure mode; D is incorrect.
10	10. Explainability of Models	B	Explainability matters most in high-stakes settings; B is incorrect.
11	11. Ashby Control	C	To control a system, the controller's variety must match (>=) the system's variety.
12	12. Viable System Model	B	Recursive, self-regulating subsystems with feedback-based oversight describe the VSM.
13	13. Class Imbalance	D	SMOTE interpolates new minority samples; accuracy is unreliable here and oversampling enlarges (not deletes) data.
14	14. Uncertainty Scenarios	A	Analysis is still possible under uncertainty; A is incorrect.
15	15. Unsupervised Learning	A	Predicting predefined labels is supervised learning; A is incorrect.
16	16. Requisite Variety	C	Requisite variety: the controller needs at least as much variety as the system/environment.
17	17. System Dynamics	B	The bullwhip effect arises from feedback and delays - a System Dynamics phenomenon.
18	18. Smart Factory	A	MES bridge physical machinery and cyber ERP for real-time control; smart factories are data-driven.
19	19. Performance Model-Evaluation	C	That definition describes precision, not recall; C is incorrect.

20	20. Unsupervised Algorithm	**A**	Clustering needs no labels; A is incorrect.
21	21. Target Centric Approach	**B**	The target-centric approach is structured and iterative, refining insights non-linearly around a goal.
22	22. Clustering & Network Analysis	**D**	Analysing connections, centrality and communities in a graph is network analysis.
23	23. Hype Myths in Data Science	**A**	~95% of big data is unstructured is true; the others are common myths.
24	24. EDA	**C**	EDA should not be skipped; C is incorrect.
25	25. Ashby Misinterpretation	**D**	Continually adding internal complexity is the misinterpretation; D is incorrect.
26	26. VUCA environment	**A**	VUCA = Volatility, Uncertainty, Complexity, Ambiguity.
27	27. Type of learner	**C**	Eager learners invest in training to build a model, then predict quickly.
28	28. Biases	**B**	Automation can scale bias; a neutral look does not prove neutrality and anonymization can fail.
29	29. VSM Early Warning	**C**	Removing feedback destroys early warning; C is incorrect.
30	30. Interdisciplinarity in Data Science	**D**	Social science (ethics, fairness) is relevant; D is incorrect.

Mock Exam 2 — Solutions & Answer Key (Single best answer (tick one) · 1 pt/question · 30 points)

Q	Topic	Answer	Rationale
1	1. VSM vs. System Dynamics	**C**	Simulating time behaviour with stocks, flows and delays is System Dynamics.
2	2. Ashby Adaptation	**B**	Adaptation aligns internal variety with the environment; freezing variety or ignoring the environment is wrong.
3	3. Data Science Distinction	**B**	Big Data and Data Science are not the same; B is incorrect.
4	4. Real-Time Adaptive Models	**A**	Concept drift is a key reason adaptive models exist; A is incorrect.
5	5. Overfitting	**C**	A large train-test gap with very high training accuracy indicates overfitting (high variance).
6	6. Visualization	**A**	Dashboards combat data overload and improve visibility, supporting trust.
7	7. Complexity Examples	**C**	Complexity = many interdependent variables with non-obvious causality; the other statements are false.
8	8. Class Weighting	**D**	Class weighting does not add/remove rows; D is incorrect.
9	9. Target-Oriented Action	**D**	The data-first start is exactly what target-orientation avoids; D is incorrect.
10	10. Explainability of Models	**B**	XAI explains the 'why' behind decisions, which builds trust and supports responsible use.
11	11. Ashby Control	**A**	CEP detects patterns in streams; SPC identifies process variations - both support control.
12	12. Viable System Model	**B**	The VSM is recursive and cybernetic with mutually regulating subsystems; the other options are false.
13	13. Class Imbalance	**C**	High accuracy can hide poor minority-class performance; C is incorrect.
14	14. Uncertainty Scenarios	**D**	Uncertainty does not mean there is no data; D is incorrect.
15	15. Unsupervised Learning	**A**	Unsupervised learning discovers structure in unlabeled data with no predefined outputs.
16	16. Requisite Variety	**C**	Fatal undercomplexity = the system's controls cannot cope with the complexity faced.
17	17. System Dynamics	**B**	Balancing loops counteract, reinforcing loops amplify; stocks are accumulations and SD handles non-linearity.
18	18. Smart Factory	**A**	A smart factory is real-time and connected, not purely offline; A is incorrect.
19	19. Performance Model-Evaluation	**B**	High training accuracy never removes the need for evaluation; B is incorrect.

20	20. Unsupervised Algorithm	**C**	K-means is an unsupervised clustering algorithm; the others are supervised.
21	21. Target Centric Approach	**B**	Organising data around the customer subject for strategic insight is target/subject-centric.
22	22. Clustering & Network Analysis	**D**	Both are exploratory; clustering groups by similarity, network analysis maps connections.
23	23. Hype Myths in Data Science	**D**	Volume does not beat quality; D is the myth and therefore incorrect.
24	24. EDA	**C**	Using test-set information during training is data leakage; C is incorrect.
25	25. Ashby Misinterpretation	**A**	Blindly adding internal complexity is a misinterpretation; attenuation is equally valid and matching variety is correct.
26	26. VUCA environment	**D**	VUCA calls for adaptive, systemic, feedback-based decision-making.
27	27. Type of learner	**D**	Lazy learners defer to query time and are memory-heavy; decision trees are eager.
28	28. Biases	**A**	Automation tends to scale bias, not remove it; A is incorrect.
29	29. VSM Early Warning	**B**	Feedback remains necessary after deployment; B is incorrect.
30	30. Interdisciplinarity in Data Science	**A**	Data Science is transdisciplinary, integrating several fields to solve complex problems.

Mock Exam 3 — Solutions & Answer Key (Single best answer (tick one) · 1 pt/question · 30 points)

Q	Topic	Answer	Rationale
1	1. VSM vs. System Dynamics	**B**	B is correct; both actually rely on feedback, SD is dynamic not static, and they are distinct approaches.
2	2. Ashby Adaptation	**B**	Reducing one's own variety is not the only (or correct) response; B is incorrect.
3	3. Data Science Distinction	**B**	Big Data and Data Science are not identical; B is incorrect.
4	4. Real-Time Adaptive Models	**D**	Real-time adaptive models update continuously from streaming feedback rather than being fixed once.
5	5. Overfitting	**A**	A shows a highly irregular boundary weaving around individual points (incl. noise) -> overfitting / high variance.
6	6. Visualization	**C**	Poor visuals can distort and imply false completeness; the other options are false.
7	7. Complexity Examples	**B**	Complexity is not the same as mere complicatedness; B is incorrect.
8	8. Class Weighting	**C**	Class weighting changes the loss, it does not duplicate data; C is incorrect.
9	9. Target-Oriented Action	**A**	Target-oriented action begins with the goal and designs everything around it.
10	10. Explainability of Models	**C**	Understanding the 'why' of a prediction requires explainability/interpretability.
11	11. Ashby Control	**D**	Control is cybernetic and requires matching variety; the other options contradict Ashby/CPS control.
12	12. Viable System Model	**D**	The VSM is not a static, feedback-free chart; D is incorrect.
13	13. Class Imbalance	**A**	Accuracy is unreliable under imbalance; A is incorrect.
14	14. Uncertainty Scenarios	**C**	Uncertainty = unknown outcomes despite data; A=complexity, D=ambiguity, B=volatility.
15	15. Unsupervised Learning	**D**	Grouping unlabeled data by similarity is unsupervised clustering.
16	16. Requisite Variety	**A**	The controller must match the system's variety; variety is distinguishable states, not data volume.
17	17. System Dynamics	**A**	System Dynamics is explicitly time-based; A is incorrect.
18	18. Smart Factory	**C**	Smart factories rely heavily on analytics; C is incorrect.
19	19. Performance Model-Evaluation	**B**	Accuracy can look high while the minority class is entirely missed; recall/precision/F1 are safer.
20	20. Unsupervised Algorithm	**A**	K-medoids uses a representative real object (medoid) as centre, which is robust to outliers.

21	21. Target Centric Approach	**B**	Target-centric means structuring everything around goals/ subjects; the other options are false.
22	22. Clustering & Network Analysis	**C**	Both are unsupervised/exploratory and do not require labels; C is incorrect.
23	23. Hype Myths in Data Science	**B**	Big Data and Data Science are not identical; B is incorrect (it is a myth, not a true statement).
24	24. EDA	**D**	EDA is the early step of understanding the data before modeling.
25	25. Ashby Misinterpretation	**B**	The correct reading is to match variety, via amplification or attenuation.
26	26. VUCA environment	**D**	Each VUCA dimension is distinct; preparation is not futile and uncertainty does not mean no data.
27	27. Type of learner	**A**	KNN is a lazy learner with slow prediction; A is incorrect.
28	28. Biases	**C**	Anonymization can still be defeated by re-identification; C is incorrect.
29	29. VSM Early Warning	**D**	Anti-oscillatory feedback loops detect and dampen emerging instability - an early-warning function.
30	30. Interdisciplinarity in Data Science	**A**	Domain knowledge is needed to frame the problem and make sense of data and results.

Mock Exam 4 — Solutions & Answer Key (Single best answer (tick one) · 1 pt/question · 30 points)

Q	Topic	Answer	Rationale
1	1. VSM vs. System Dynamics	**C**	Both are feedback-based systems-thinking approaches; C is the incorrect statement.
2	2. Ashby Adaptation	**B**	When environmental variety exceeds capacity the environment dominates; B is incorrect.
3	3. Data Science Distinction	**A**	Big Data = data traits/infrastructure (4 Vs); Data Science = methods + theory + application across disciplines.
4	4. Real-Time Adaptive Models	**C**	Continuous adjustment from live feedback is a closed-loop, real-time adaptive model.
5	5. Overfitting	**B**	Overfitting = high variance (learns noise); underfitting = high bias (too simple).
6	6. Visualization	**D**	Interpreting visualizations requires visual literacy; D is incorrect.
7	7. Complexity Examples	**C**	Full overview/control of complex systems is not possible; C is incorrect.
8	8. Class Weighting	**B**	Class weighting adjusts the loss to value the minority class more, leaving the data unchanged.
9	9. Target-Oriented Action	**D**	Actionable knowledge can be measured and acted upon.
10	10. Explainability of Models	**C**	Good XAI is interactive and user-specific; opacity reduces trust and large models are hard to interpret.
11	11. Ashby Control	**D**	A low-variety controller cannot manage a high-variety system; D is incorrect.
12	12. Viable System Model	**B**	The VSM is recursive, not a linear plan; B is incorrect.
13	13. Class Imbalance	**C**	Under imbalance, a trivial majority-class predictor looks accurate yet misses the minority class.
14	14. Uncertainty Scenarios	**B**	Unknown outcomes despite available data is uncertainty.
15	15. Unsupervised Learning	**A**	No labels => cheaper data but harder evaluation; the other statements are false.
16	16. Requisite Variety	**D**	Requisite variety is about control capacity, not raw data volume; D is incorrect.
17	17. System Dynamics	**D**	Reinforcing loops amplify rather than stabilise; D is incorrect.
18	18. Smart Factory	**A**	A smart factory is a CPS-based connected production system - the core of Industry 4.0.
19	19. Performance Model-Evaluation	**A**	A shows an overly simple (straight-line) boundary that misclassifies many points -> underfitting / high bias.

20	20. Unsupervised Algorithm	**B**	k is set beforehand for K-means; clustering needs no labels, Fuzzy C-means is soft, K-medoids uses a real object.
21	21. Target Centric Approach	**A**	Target-centric means goal-first, not data-first; A is incorrect.
22	22. Clustering & Network Analysis	**D**	Clustering is unsupervised and needs no labels; D is incorrect.
23	23. Hype Myths in Data Science	**C**	'More data is always better' is the myth; relevance and quality matter more than sheer volume.
24	24. EDA	**A**	EDA findings should be documented and handled (imputation/cleaning) before modeling.
25	25. Ashby Misinterpretation	**D**	Matching variety via amplification or attenuation is correct; the others are misinterpretations.
26	26. VUCA environment	**C**	The four VUCA dimensions are distinct; C is incorrect.
27	27. Type of learner	**A**	Lazy learners predict slowly because they search the data; A is incorrect.
28	28. Biases	**A**	Sampling bias = unrepresentative sample. B=confirmation, C=survivorship, D=automation bias.
29	29. VSM Early Warning	**B**	Rising variation on SPC charts is an early-warning signal to intervene before defects occur.
30	30. Interdisciplinarity in Data Science	**B**	Transdisciplinarity means genuine integration; algorithms-only and ignoring domain/social science are wrong.

Mock Exam 5 — Solutions & Answer Key (Single best answer (tick one) · 1 pt/question · 30 points)

Q	Topic	Answer	Rationale
1	1. VSM vs. System Dynamics	**D**	Both are built on feedback; D is the incorrect statement.
2	2. Ashby Adaptation	**C**	Adaptation = aligning internal variety with environmental variety, via amplification or attenuation.
3	3. Data Science Distinction	**C**	DIKIW = Data, Information, Knowledge, Intelligence, Wisdom.
4	4. Real-Time Adaptive Models	**C**	Model-free RL learns from rewards; feedback is essential and concept drift is exactly why adaptation matters.
5	5. Overfitting	**C**	Increasing complexity worsens overfitting; C is incorrect.
6	6. Visualization	**B**	Visualizations are vulnerable to designer bias; B is incorrect.
7	7. Complexity Examples	**B**	Complex CPS data is large, high-frequency, heterogeneous and non-IID.
8	8. Class Weighting	**B**	Class weighting changes the loss; it does not modify the dataset like sampling/SMOTE.
9	9. Target-Oriented Action	**B**	Targets should be measurable with aligned stakeholders; the other options are anti-patterns.
10	10. Explainability of Models	**A**	Opaque models are not ideal for high-stakes decisions; A is incorrect.
11	11. Ashby Control	**B**	Requisite variety forbids controlling a higher-variety system with a lower-variety controller; B is incorrect.
12	12. Viable System Model	**C**	The VSM is a recursive management structure with anti-oscillatory feedback ensuring viability.
13	13. Class Imbalance	**D**	Fraud/rare-event detection with a tiny positive class is the classic imbalance case.
14	14. Uncertainty Scenarios	**B**	Scenario thinking helps under uncertainty; the other statements are false.
15	15. Unsupervised Learning	**D**	Unsupervised learning needs no labels; D is incorrect.
16	16. Requisite Variety	**A**	Data alone does not guarantee control; A is incorrect.
17	17. System Dynamics	**A**	System Dynamics models time behaviour using stocks, flows, feedback loops and delays.
18	18. Smart Factory	**B**	Forecasting failures from sensor data to act in advance is predictive maintenance.
19	19. Performance Model-Evaluation	**A**	F1 balances precision and recall. The diagonal is random; the confusion matrix is for classification.
20	20. Unsupervised Algorithm	**D**	Fuzzy C-means allows partial (soft) membership; D is incorrect.

21	21. Target Centric Approach	**A**	It is the opposite of a generic system-centered approach; A is incorrect.
22	22. Clustering & Network Analysis	**D**	Clustering = grouping by similarity; network analysis = studying relationships/connections.
23	23. Hype Myths in Data Science	**A**	Lacking metadata/governance, a data lake degenerates into an unusable data swamp.
24	24. EDA	**C**	EDA examines the data to guide modeling; using test info is leakage and EDA is never skippable.
25	25. Ashby Misinterpretation	**D**	Equating variety with data volume is a misinterpretation; D is incorrect.
26	26. VUCA environment	**A**	Assuming preparation is futile is a misconception; A is incorrect.
27	27. Type of learner	**D**	KNN is the classic lazy (instance-based) learner; the others are eager learners.
28	28. Biases	**A**	Historic inequities in the data are learned and amplified by the model (algorithmic bias).
29	29. VSM Early Warning	**B**	Feedback enables catching deviations early; removing feedback destroys early warning.
30	30. Interdisciplinarity in Data Science	**C**	Data Science is transdisciplinary, not purely software engineering; C is incorrect.